

《GestureControl Workshop Step-by-Step Guide (For Teachers)》

Final PDF-ready text with screenshot placement instructions

1. Workshop Overview

This workshop demonstrates how AI-powered hand tracking can support learning sign language.

Students will use their webcam, perform hand gestures, and observe real-time recognition powered by MediaPipe and a custom backend.

2. Requirements

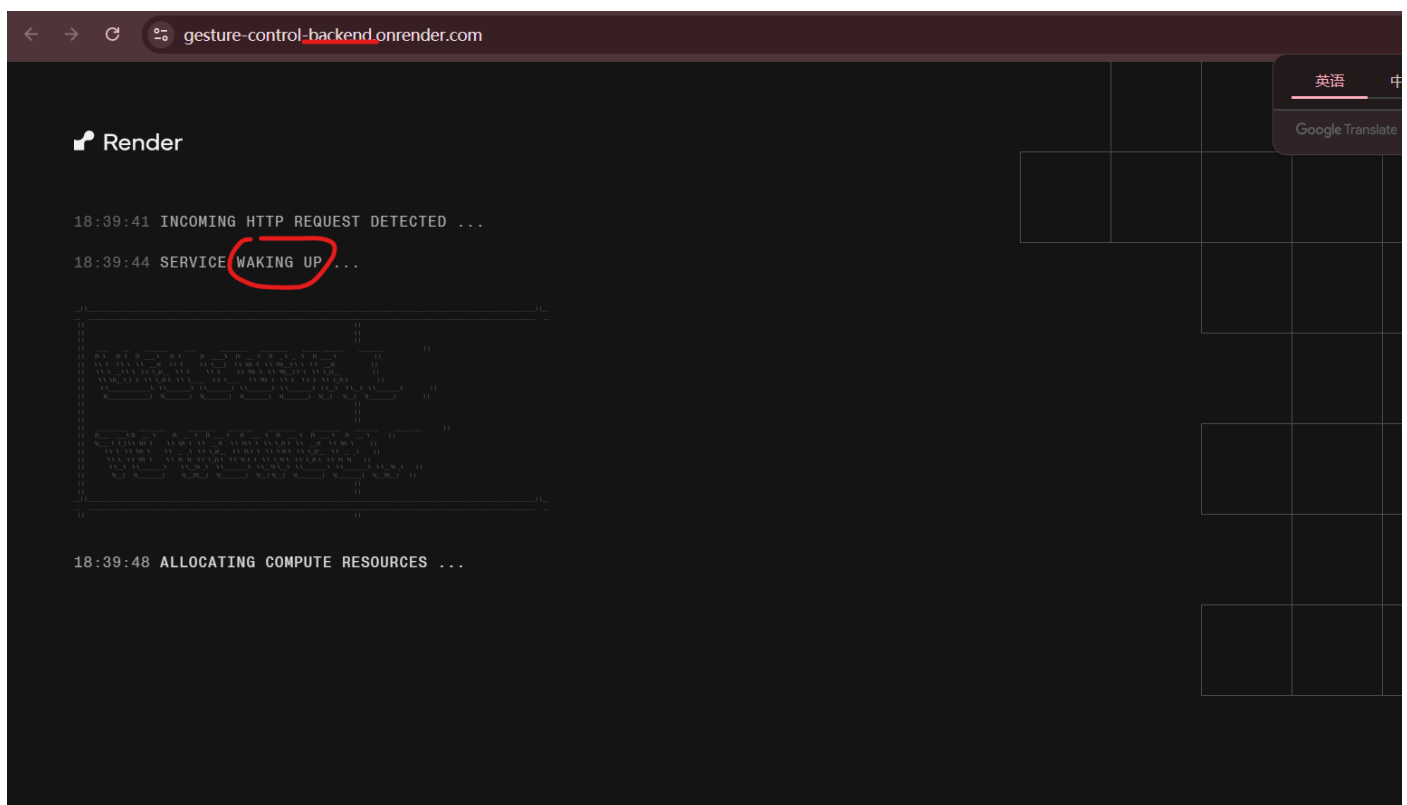
- Laptop with webcam
- Google Chrome (recommended)
- Internet connection
- Access to both project URLs:
 - Backend: <https://gesture-control-backend.onrender.com>
 - Frontend: <https://gesture-control-xli6.onrender.com>

3. Step-by-Step Instructions

Step 1 — Activate the Backend Service

Before running the workshop, the backend must be woken up from sleep (Render free tier auto-sleep behavior).

1. Open the backend URL:
<https://gesture-control-backend.onrender.com>
2. Wait for the page to show messages such as:
 - a. “SERVICE WAKING UP...”
 - b. “ALLOCATING COMPUTE RESOURCES...”
3. Once the logs stop scrolling, the backend is ready.

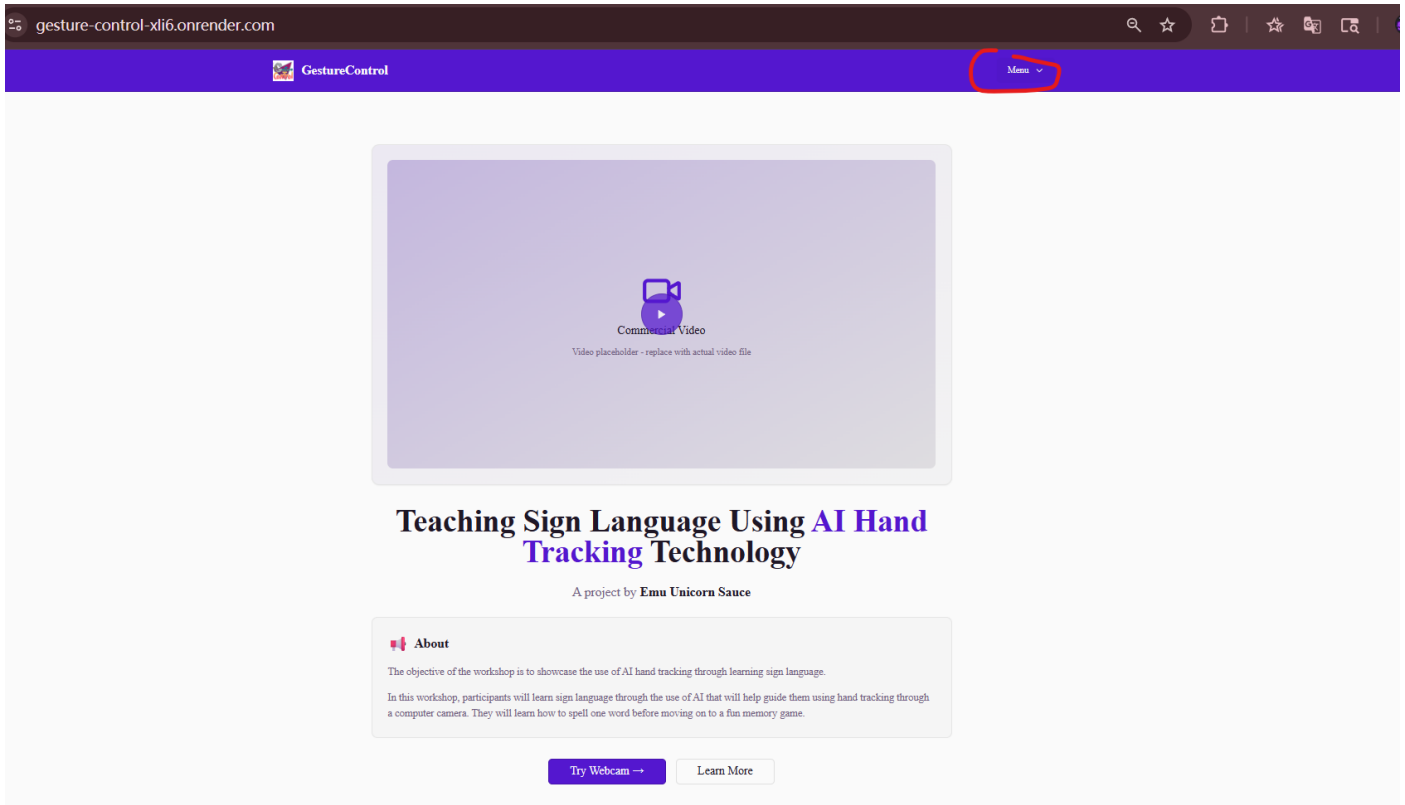


Step 2 — Open the Frontend Website

After the backend is active, open the frontend site:

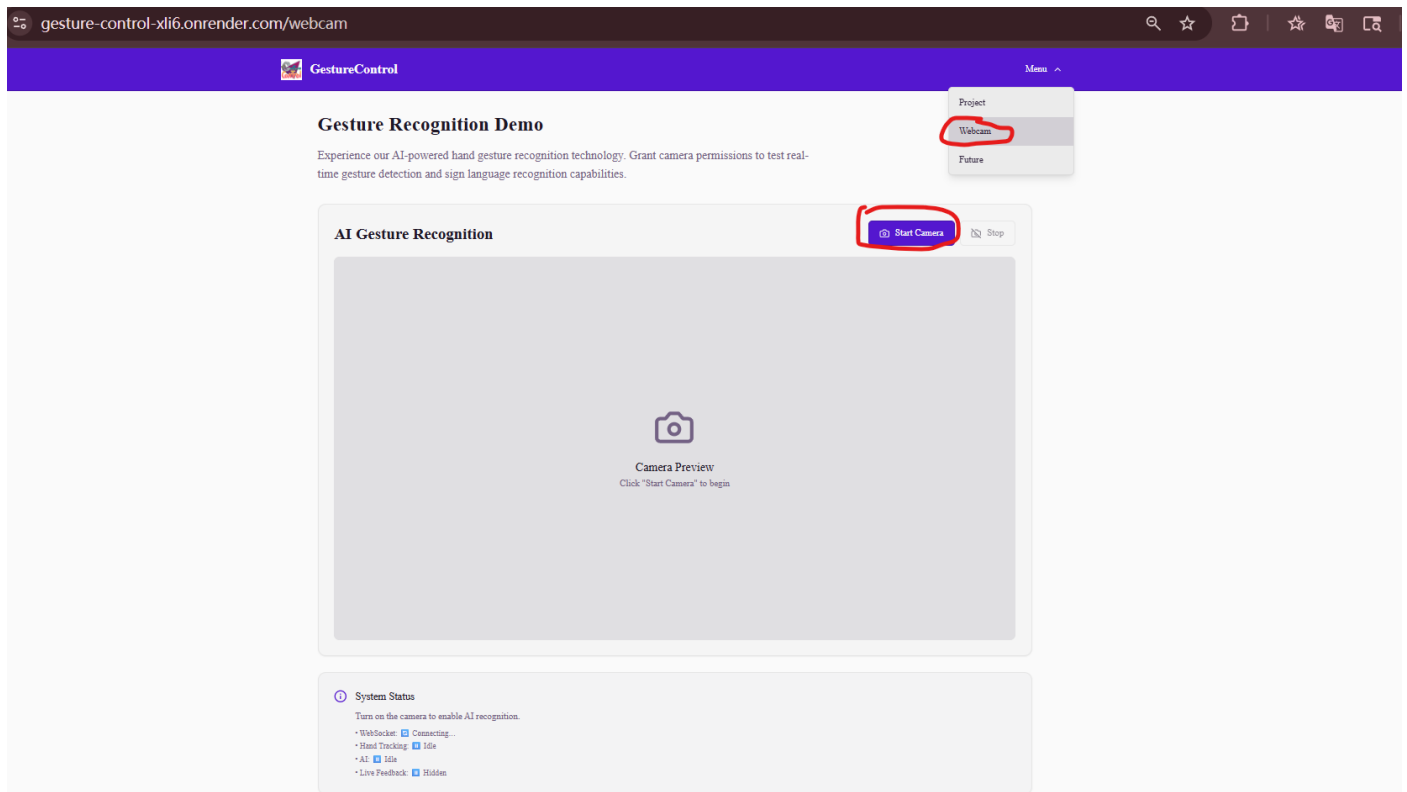
<https://gesture-control-xli6.onrender.com>

You should see the homepage displaying the header, video placeholder, and navigation menu.



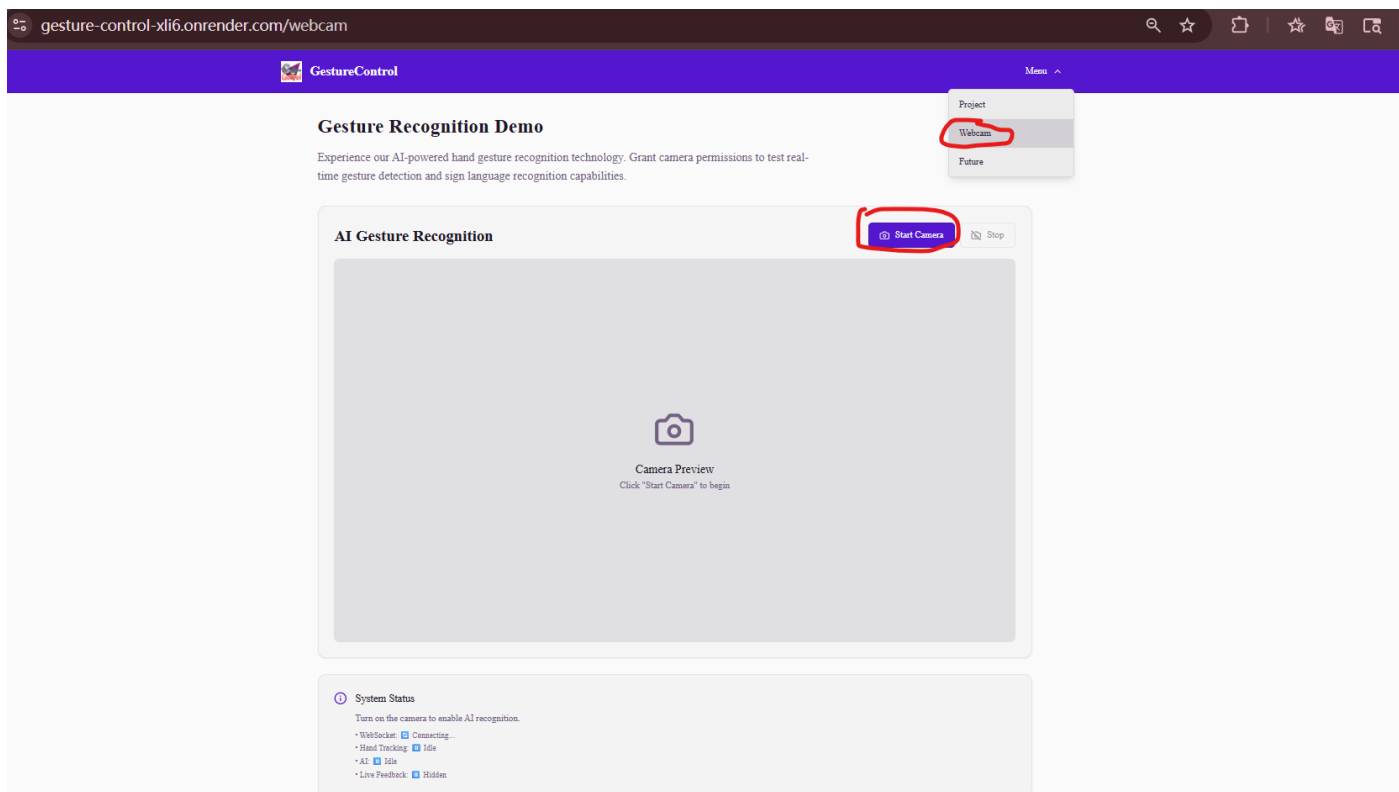
Step 3 — Navigate to the Webcam Demo

1. Click the "Menu" button located in the upper right corner.
2. Select "Webcam" from the dropdown options.
This will take you to the gesture recognition demo page.



Step 4 — Start the Camera

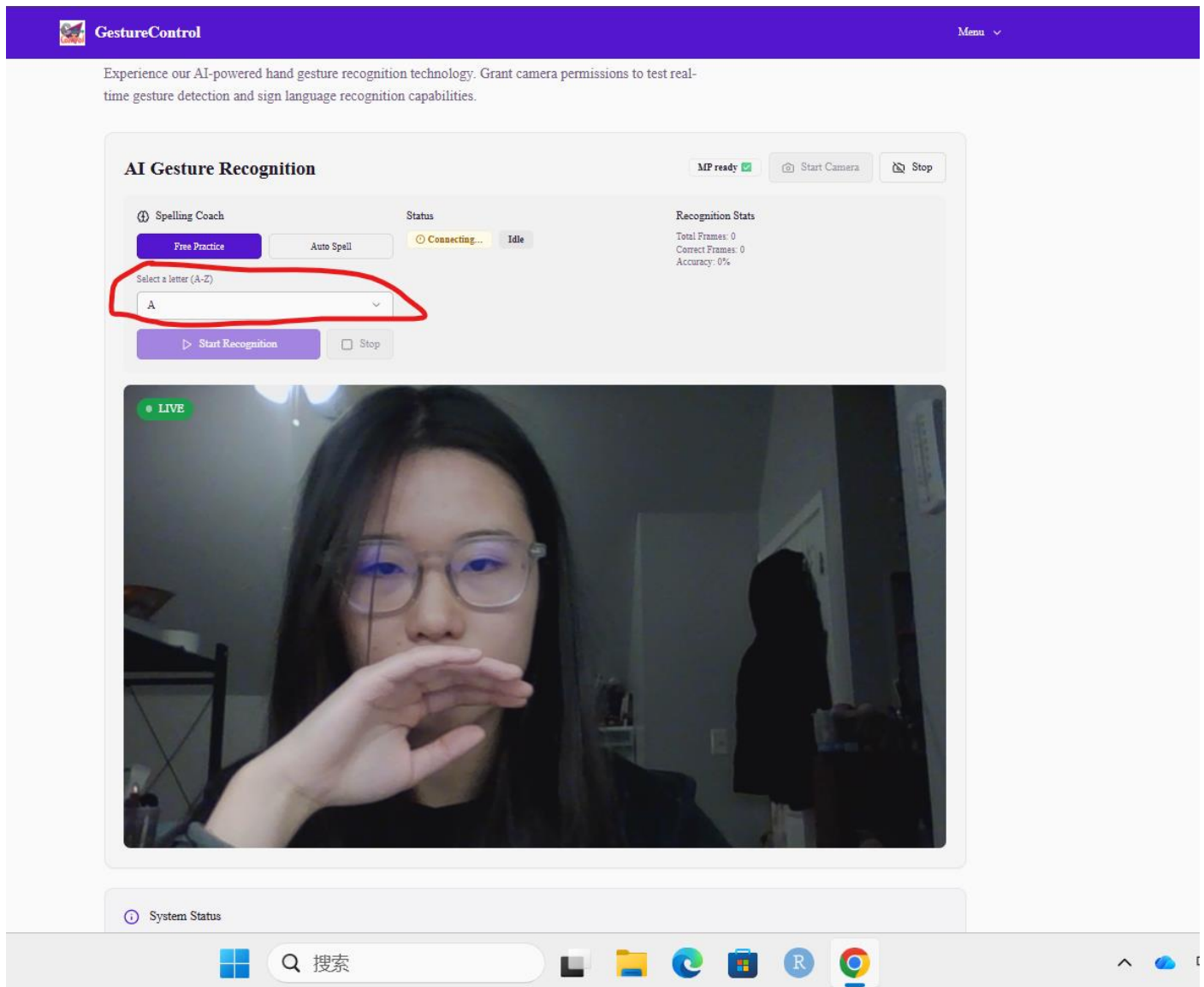
1. Click the "Start Camera" button.
2. When the browser prompts for camera permission, click "Allow."
3. A live video preview should appear in the camera window.



Step 5 — Select a Letter to Practice

Use the dropdown menu labeled “Select a letter (A–Z).”

Choose any letter, such as A, B, or C, depending on the demonstration.

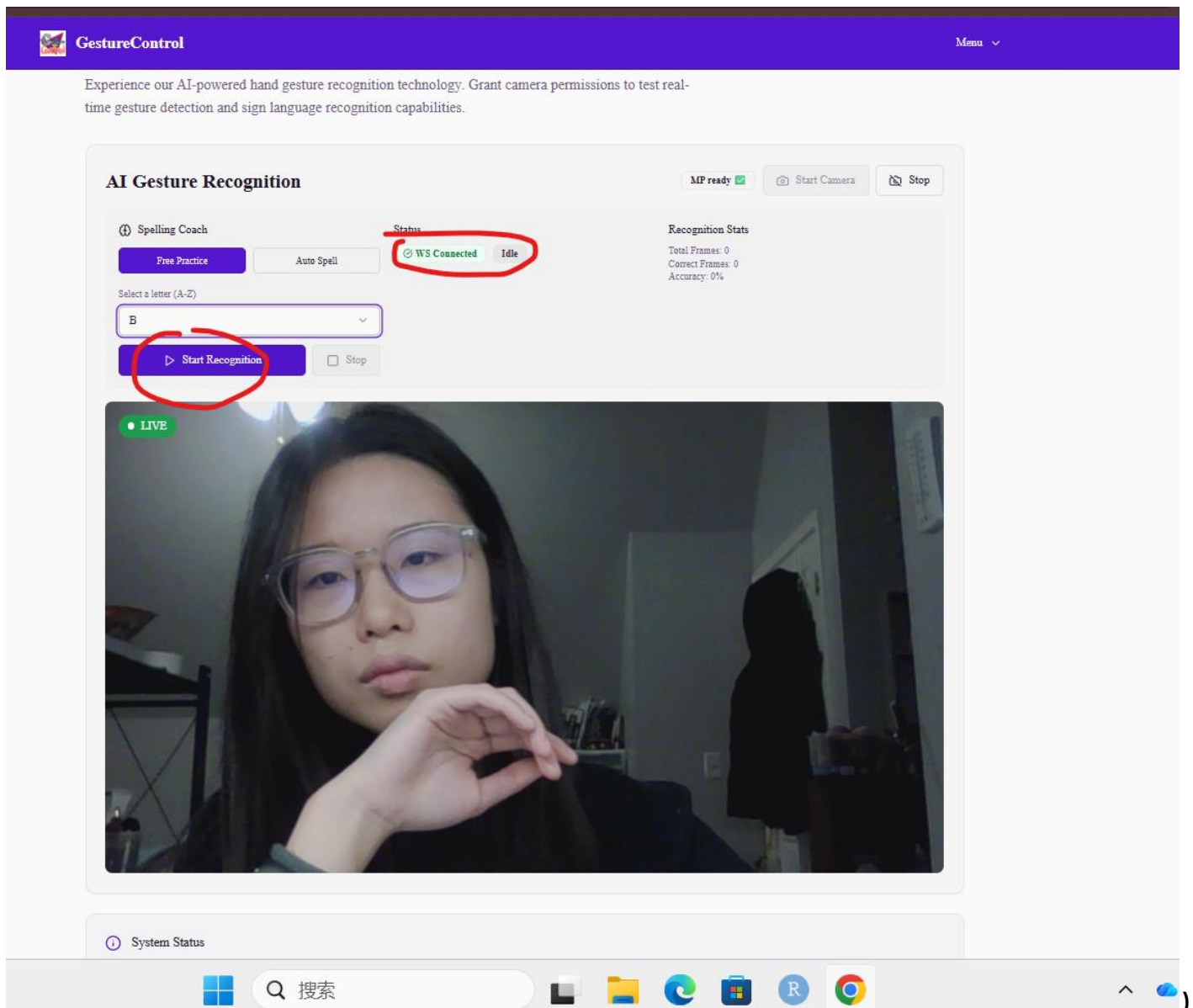


Step 6 — Confirm Backend Connection and Start Recognition

Before starting recognition, check that the system status shows:

- "WS Connected" (WebSocket connection active)
- "Idle" or "Ready"

Once connected, click "Start Recognition."



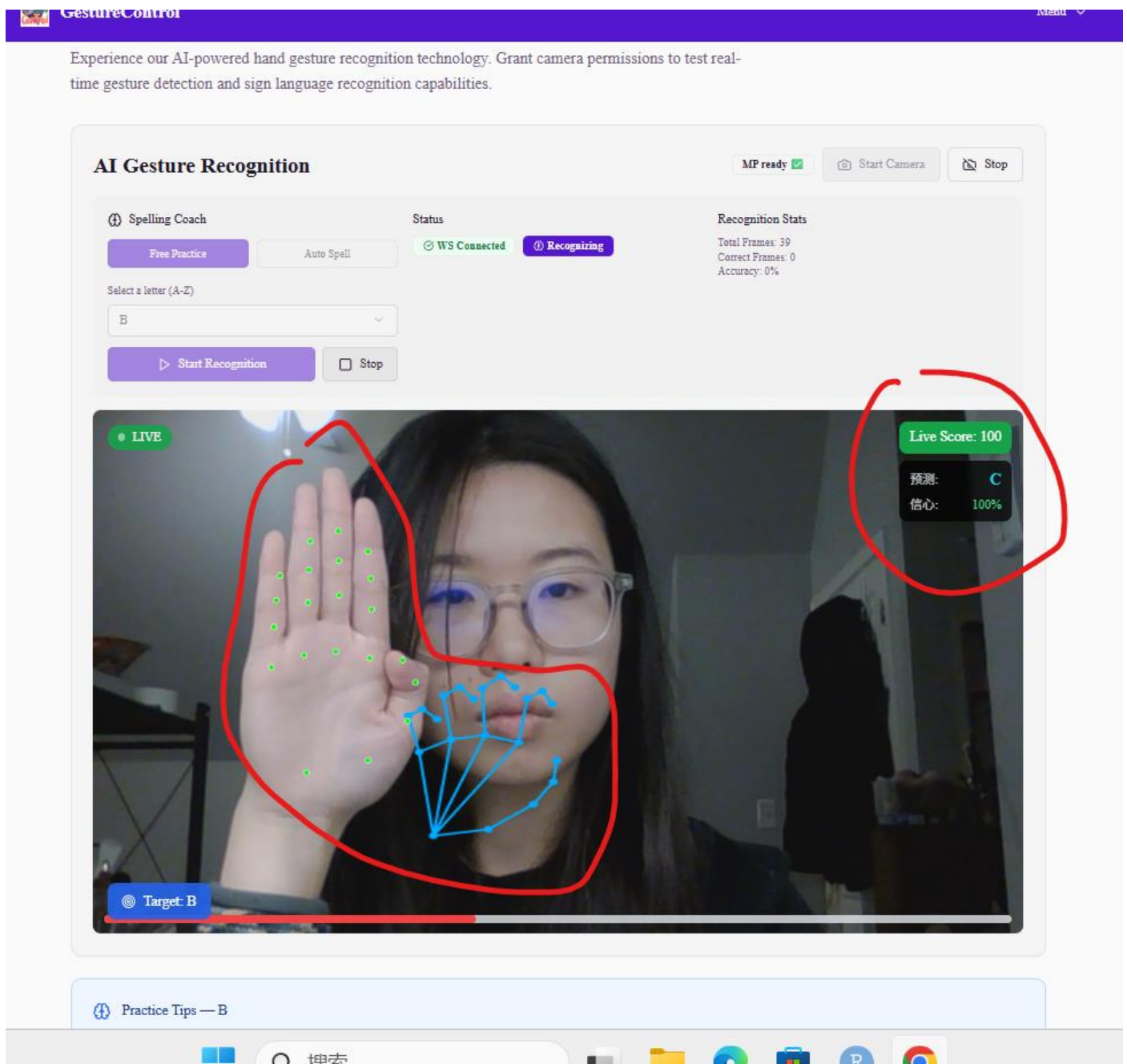
Step 7 — Perform a Hand Gesture

Ask students to perform the ASL gesture corresponding to the selected letter.

The system will display:

- Detected hand landmarks
- Gesture prediction
- Confidence score
- Live score or recognition feedback

This step demonstrates real-time gesture recognition and forms the core part of the workshop exercise.



4. Classroom Tips for Teachers

- Ensure adequate lighting for the camera to detect hand landmarks clearly.
- Have students perform gestures one at a time to avoid multi-hand confusion.
- Ask students to hold their gesture still for one to two seconds for best results.
- If recognition is inconsistent, adjust hand distance or angle relative to the camera.

5. Common Issues and Quick Fixes

Issue	Solution
Camera feed does not appear	Refresh page, click “Start Camera,” and allow permission
Backend not responding	Re-open backend URL and wait for the “waking up” sequence to finish
Recognition seems inaccurate	Improve lighting, raise hand higher, reduce background noise
WS not connected	Refresh both backend and frontend pages again

6. Suggested Workshop Structure

- 5 min introduction to AI hand tracking
- 5 min backend + frontend setup
- 10 min students practicing gesture recognition
- 10 min spelling or motion activity
- 5 min class discussion and Q&A